
Hospital Patient Data Overview

Project Report:

Project Title: Hospital Patient Data Overview

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Project Type: Exploratory Data Analysis (EDA) with Tkinter Dashboard

Programming Language: Python

Libraries Used: Pandas, NumPy, Matplotlib, Tkinter, Jupyter Notebook

1. Introduction:

Healthcare organizations generate large amounts of patient data every day. Proper analysis of this data helps hospitals improve patient care, understand disease trends, and make better management decisions. This project focuses on analyzing hospital patient records using Python and Exploratory Data Analysis (EDA). The project also includes a graphical user interface (GUI) developed with Tkinter for displaying key information and visualizations.

2. Project Objectives:

The main objectives of this project are:

- Analyze patient demographic information.
 - Identify the most common diseases among patients.
 - Study different admission types.
 - Evaluate treatment outcomes.
 - Analyze hospital bills.
 - Examine the average length of patient stay.
 - Present the analysis through graphs and a simple Tkinter dashboard.
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3. Dataset Description:

The dataset contains hospital patient records with the following attributes:

Column	Description
Patient_ID	Unique patient identification number
Age	Patient age
Gender	Male or Female
Disease	Diagnosed disease
Admission_Type	Emergency or Routine admission
Admission_Date	Date of hospital admission
Discharge_Date	Date of hospital discharge
Length_of_Stay	Total days spent in hospital
Treatment	Treatment provided
Outcome	Recovery status
Hospital_Bill	Total treatment cost

4. Software and Tools Used:

- Python
 - Visual Studio Code
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Tkinter
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5. Data Preprocessing:

Before analysis, the dataset was cleaned using Python.

The following preprocessing steps were performed:

- Removed duplicate records.
 - Checked and handled missing values.
 - Converted date columns into datetime format.
 - Verified data types.
 - Saved the cleaned dataset as **cleaned_data.csv**.
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6. Exploratory Data Analysis (EDA):

Several visualizations were created to understand the dataset.

Age Distribution:

A histogram was used to observe the age distribution of patients. The graph shows that patients belong to different age groups, with middle-aged adults forming the majority.

Gender Distribution:

A bar chart compares the number of male and female patients admitted to the hospital.

Disease Distribution:

A bar chart identifies the most common diseases among patients.

Treatment Outcomes:

A pie chart illustrates the percentage of recovered patients, patients still under treatment, and other outcomes.

Hospital Bill Distribution:

A histogram shows how treatment costs vary among patients.

Length of Stay:

Another histogram illustrates how many days patients stayed in the hospital.

7. Statistical Analysis:

The project calculated several useful statistics, including:

- Average patient age
- Average hospital bill
- Maximum hospital bill
- Minimum hospital bill
- Average length of hospital stay

These statistics provide a quick overview of hospital performance and patient characteristics.

8. Tkinter Dashboard:

A graphical user interface (GUI) was developed using Tkinter.

The dashboard includes the following options:

- Patient Summary
- Age Distribution
- Gender Distribution
- Disease Distribution
- Treatment Outcome Analysis
- Exit Button

The dashboard allows users to visualize patient information with a simple interface.

9. Results:

The analysis produced several valuable insights:

- Most patients belong to the middle-age category.
 - Emergency admissions are more common than routine admissions.
 - Certain diseases occur more frequently than others.
 - Most patients recovered successfully after treatment.
 - Hospital bills increase with longer hospital stays.
 - Data visualization makes patient information easier to understand.
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10. Challenges:

During project development, the following challenges were encountered:

- Handling missing values.
 - Cleaning inconsistent records.
 - Selecting appropriate visualizations.
 - Managing file paths between folders.
 - Integrating Matplotlib graphs with Tkinter.
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11. Future Improvements:

The project can be enhanced by adding:

- Real-time hospital database connectivity.
 - Machine Learning models for disease prediction.
 - Interactive charts using Plotly.
 - User authentication.
 - Export reports in PDF and Excel formats.
 - Advanced hospital management features.
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12. Conclusion:

This project demonstrates how Python can be used for healthcare data analysis through Exploratory Data Analysis (EDA). Using Pandas for data processing, Matplotlib for visualization, and Tkinter for the graphical interface, the project provides meaningful insights into patient demographics, diseases, treatment outcomes, hospital expenses, and length of stay. The developed system is simple, efficient, and can serve as a foundation for more advanced hospital analytics applications.

References:

1. Python Software Foundation. *Python Documentation*. <https://docs.python.org/>
2. Pandas Documentation. <https://pandas.pydata.org/>
3. Matplotlib Documentation. <https://matplotlib.org/>
4. NumPy Documentation. <https://numpy.org/>
5. Tkinter Documentation. <https://docs.python.org/3/library/tkinter.html>